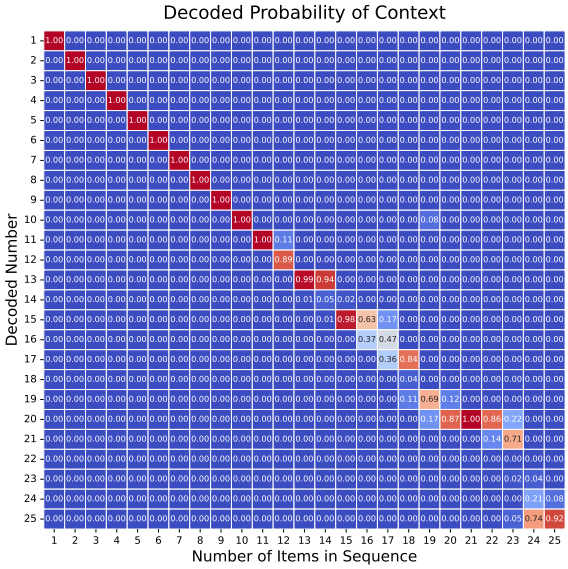



Figure 3: Decoded output probabilities for the unstructured baseline method on Qwen2.5 7B. The heatmap shows the decoded probabilities of model outputs, averaged over different item types, for target counts ranging from 1 to 25. As the count increases beyond 10, the diagonal entries gradually fade, indicating reduced model confidence.

reported in Tables 1 and 2. Across all models, we find that only one configuration consistently succeeds on long contexts: structured inputs combined with intermediate reasoning steps.

Surprisingly, encouraging CoT reasoning alone, without structured input, does not yield a notable improvement. Furthermore, structured input without intermediate steps is also ineffective and can be harmful in some cases. A likely explanation is that models must first consolidate partial results in intermediate steps before aggregating them into the final answer through a two-stage process. Our observational and causal analyses provide further insight into why models cannot simultaneously gather partial counts from the context and add them up without loss. Overall, these results show that neither external structure nor reasoning alone is sufficient. Their combination is necessary to overcome large-scale counting failures.

## 4 Attention Analysis

We analyze attention patterns of Qwen2.5 7B (Yang et al., 2024a) to understand how the proposed System-2 strategy enables large-scale counting. We first identify which tokens are attended to when the model generates (i) intermediate partition-level counts and (ii) the final aggregated answer. We then localize the layers and heads that contribute most to these processes.

Figure 4a shows attention patterns to the input items when the model generates intermediate reasoning steps. We consider a structured context with three partitions containing 3, 4, and 5 items. During the generation of a reasoning token such as "part 2: 4", the attention weight of the generated number peaks sharply on the final item and the final comma separator of the corresponding partition. This pattern is consistent across prompts and is most pronounced in middle to late layers, typically around layers 19 to 23. Earlier items within the same partition receive substantially lower attention.

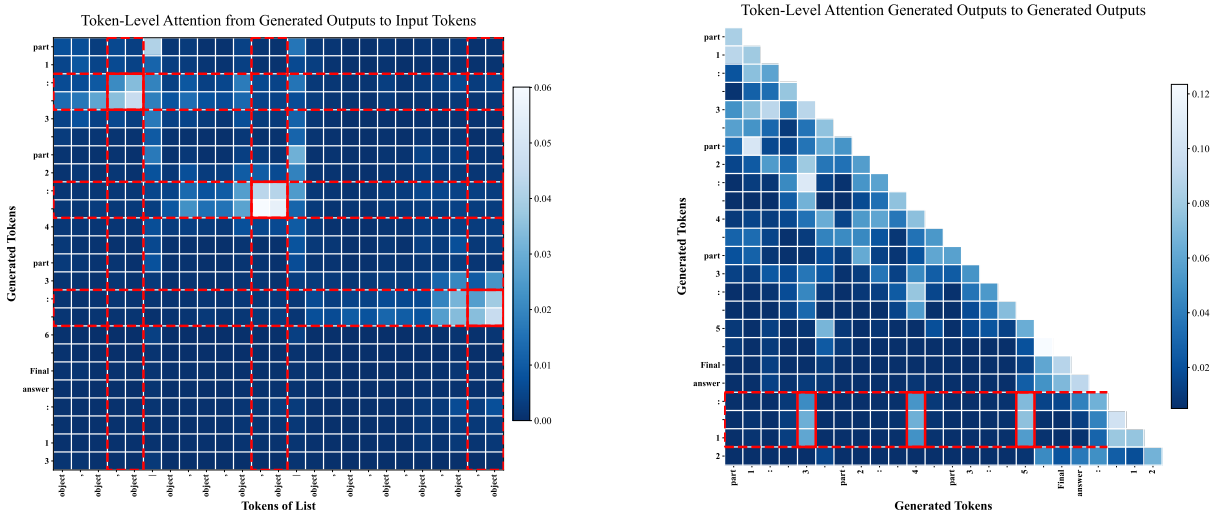
Figure 4b shows attention patterns when the model generates the final answer. The ":" token and the whitespace token immediately preceding the final number attend strongly to all intermediate reasoning numbers produced earlier (e.g., 3, 4, and 5). The attention mass is distributed across these tokens, with a clear concentration in the same layers identified for intermediate steps. Attention to the original input items is weak at this stage.

To further localize the attention pathways involved in intermediate reasoning and final aggregation, we focus on the most relevant tokens identified in Figure 4. We average attention values across different configurations, including item types and partition sizes. Figure 5 shows that attention peaks in layers 19 to 23, highlighting the central role of these layers in information transfer and aggregation. Figure 6 further shows that the most influential heads are concentrated in layers 21, 22, and 23. For example, head 13 in layer 22 consistently exhibits high attention to the selected tokens.

Together, these results suggest a staged computation. First, each partition is counted independently, with the final tokens of the partition encoding the local count. Second, these local counts are written into intermediate reasoning tokens. Finally, the model attends to these intermediate tokens and aggregates their values to produce the final answer. These observations are consistent with a hypothetical mechanism that separates counting, information transfer, and aggregation into distinct components.

## 5 Causal Mediation Analysis

To assess the hypothesis of a multi-stage System-2 counting mechanism suggested by the attention analysis, we perform a set of causal mediation experiments based on activation patching (Heimersheim and Nanda, 2024; Zhang and Nanda, 2024),




(a) Attention of generated tokens to input tokens.

(b) Attention of generated tokens to generated tokens.

Figure 4: Attention patterns of selected tokens under the System-2 counting strategy. Attention values are averaged across layers 19 to 23, all heads, and all item types (e.g., different fruit names).

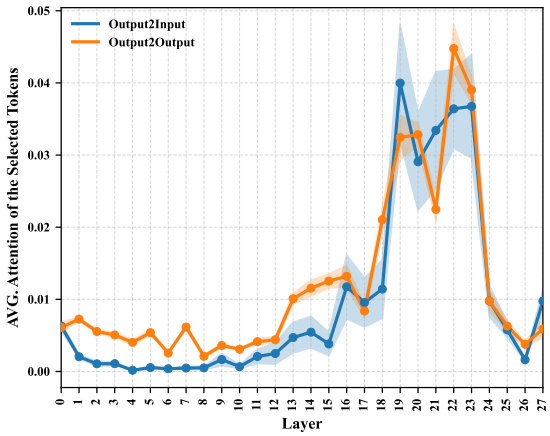



Figure 5: Average attention of correctly generated intermediate counts to the final item of their partition, alongside attention of the final answer to partition-level counts, across layers. Higher attention values are observed from layer 19 to 23 for both paths.

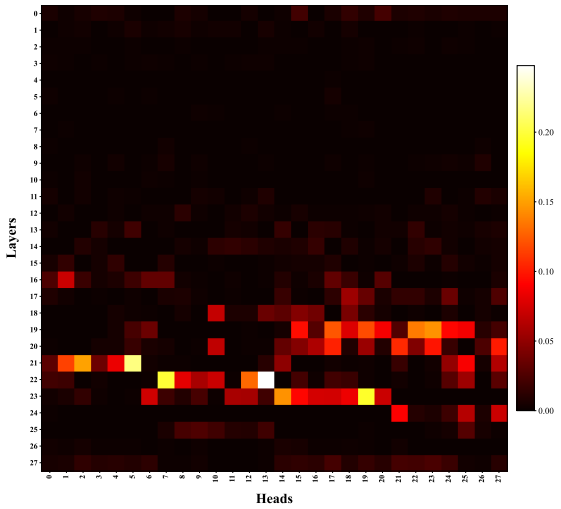



Figure 6: Heatmap of average attention to final items of partitions across layers and heads for intermediate counting steps. Layers and heads responsible for transferring numerical information from input partitions to intermediate counts appear with higher intensity.

masking ablation, and attention knockout (Geva et al., 2023). The goal is to identify where partition-level count information is stored, how it is transferred and consolidated, and which components are causally required for correct aggregation.

### 5.1 Token-Level Information Probing

We first examine where partition-level counts are encoded in the input context. This analysis requires vocabulary projection and probing tools. We observe that existing tools, such as logit-lens (NostAlgebraist, 2020) and tuned-lens (Belrose et al., 2023), are not reliable for decoding numerical information. We therefore use the CountScope method (Hasani et al., 2025b) to decode the implicit count associated with tokens. CountScope is a causal probing method that patches the activation of a target token into a blank counting context and decodes the model's output as the implied count.

Figure 7 shows the average decoding probability of ground-truth numbers across different token types. Ground-truth numbers correspond to the number of items in each partition. The results show that the ground-truth value is encoded with high confidence primarily at the final item and the final comma separator of each partition. For the first partition, the confidence at the final item is high; however, for subsequent partitions, the confidence

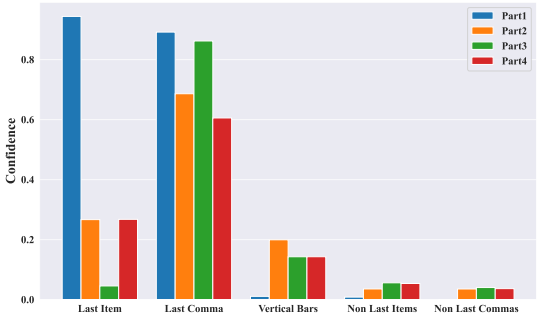



Figure 7: Ground-truth probabilities across different tokens of partitions decoded by CountScope. Probabilities are averaged over different item types and configurations (3 to 9 items per partition).

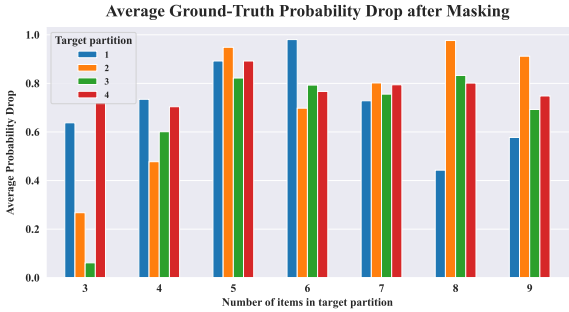



Figure 8: Average ground-truth probability drop after masking the final item and comma (e.g., ‘, apple’) from each partition, showing the effect on the target count for digit sizes ranging from 3 to 9.

at this token decreases, while the confidence at the vertical bar character increases. Nevertheless, the final comma separator reliably stores the latent count of the corresponding partition. These findings are consistent with the attention results in Section 4, which showed that decoded numbers of intermediate steps attend most strongly to these final partition tokens.

More interestingly, this experiment reveals that at each partition boundary, the count resets and begins again. As a result, the final item of each partition encodes the number of items counted since the beginning of that partition. This observation explains why the System-2 strategy avoids the large-number failure observed in unstructured inputs. Each partition is counted independently, and its size remains within the range where the model’s implicit counter is accurate.

### 5.2 Token-Level Causal Interventions

To test whether the identified tokens causally mediate the model’s behavior, we perform zero ablation on the final items and separators of each partition. Specifically, after processing the entire input sequence and extracting token embeddings, we replace the activation values of selected tokens with zeros across all layers.

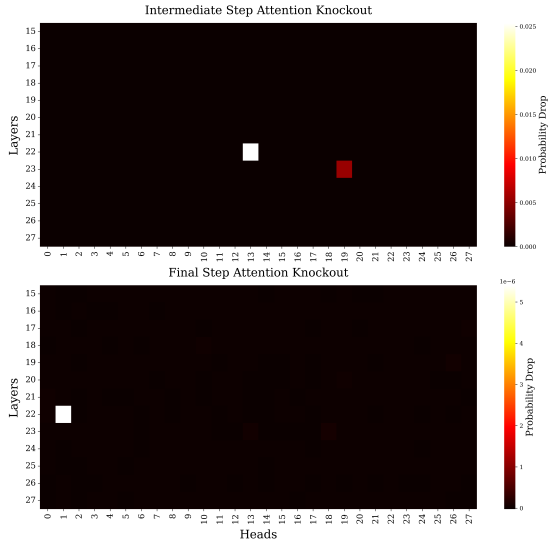



Figure 9: Heatmap of the average probability drop after attention knockout across layers and heads for intermediate (top) and final counting steps (bottom). Attention knockout is applied only to the identified effective tokens from the input context and intermediate steps.

Figure 8 shows the average probability drop after zero ablation of the final item and final comma separator. This intervention leads to a sharp drop in the probability of generating the correct partition-level count in the corresponding reasoning step. For example, ablating the final “, apple” token of a four-item partition significantly reduces the likelihood of generating the token “4”.

### 5.3 Information Pathway Localization

To perform fine-grained circuit localization, we analyze the pathways responsible for System-2 counting using attention knockout (Geva et al., 2023). We selectively block individual attention heads and layers and measure the resulting drop in counting accuracy for intermediate and final steps. Figure 9 shows that for Qwen2.5 7B, the most influential components are concentrated in layer 22. In particular, attention head 13 is most important for intermediate steps, while head 1 is most important for the final aggregation step.

Notably, head 13 in layer 22 also exhibits the highest attention values in the attention heatmap shown in Figure 6. Compared to Figure 6, the pattern in Figure 9 is considerably sparser. A plausible explanation is the presence of parallel information pathways, where knocking out a single pathway does not fully disrupt the computation because